

Inter-area Oscillation Damping Controller for DFIG based Wind Power Plants

Sami Alalwani

*Department of Electrical and Computer
Engineering*

*FREEDM System Center,
North Carolina State University
Raleigh, NC, USA
shalalwa@ncsu.edu*

Semih Isik

*Department of Electrical and Computer
Engineering*

*FREEDM System Center,
North Carolina State University
Raleigh, NC, USA
sistik@ncsu.edu*

Subhashish Bhattacharya

*Department of Electrical and Computer
Engineering*

*FREEDM System Center,
North Carolina State University
Raleigh, NC, USA
sbhatta4@ncsu.edu*

Abstract —Low-frequency inter-area oscillation is typical for transmission tie-line interconnecting neighboring power systems. Inter-area modes result from generators in one area of the power system oscillating against another group of generators in another area across a weak transmission tie line. Modulating active and reactive power in a wind turbine system equipped with a Doubly Fed Induction Generator (DFIG) may effectively dampen inter-area oscillation. In this paper, modifications of Rotor Side Converter (RSC) control and Grid Side Converter (GSC) control by adding active and reactive Power Oscillation Damping (POD) controllers to improve the dynamic system response are studied. Kundur's well-known two-area system is adopted to demonstrate the impacts of active POD and reactive POD controllers of DFIG on the inter-area oscillation using the PSCAD electromagnetic simulation tool. Besides, system stability is studied based on a frequency sweep using the impedance ratio of the different subsystems inside Kundur's system. The Nyquist Stability Criterion (NSC) is adopted to determine system stability.

Index Terms— DFIG, GSC, RSC, NSC, Inter-area, Oscillation, Damping, Wind power.

I. INTRODUCTION

Recently, unprecedented growth in wind power installation has been observed due to the need for more environmentally friendly, clean power generation sources. It is predicted that demand for wind energy sources will increase in the following decades to reduce the energy shortage and environmental pressure [1]. Doubly fed induction generator (DFIG) has a significant market share as compared to other types of wind power generators due to its relatively more minor in size, low power rating, ability to operate within $\pm 33\%$ of synchronous speed, independent active and reactive power control, and high efficiency [2]. Thus, deploying more DFIG-based wind farms will significantly participate in the operation of future power grids. On the other hand, as penetration of inverter-based energy resources increases, new challenges to the grid are rising, such as reducing system stability margin, short circuit strength and system inertia. Inter-area power oscillations occur between multi-machine power systems, typically observed in weak, heavy-loaded tie-line [3]. They can affect the power system stability and voltage regulation. Typically, low-

frequency oscillations are below 2 Hz, triggered mainly by the relative swing between the rotors of parallel synchronous generators. These oscillations are mainly caused by minor disturbances in the system such as sudden changes in load, line faults, or generator outputs [4]. Interarea power oscillations are traditionally damped by reactive power modulation using Static Var Compensator (SVC) and Static Synchronous Compensator (STATCOM) [5],[6]. A Power System Stabilizer (PSS) attached to a synchronous generator is usually used for local low-frequency oscillation damping. However, they suffer from limited ability to damp interarea oscillations mode. DFIG can participate in low-power oscillatory modes by displacing synchronous machines with or without PSS [7]-[10]. It is shown in [9] that variable speed wind turbines improve the damping of modes of inter-area oscillations while worsening other modes using eigenvalue sensitivity analysis. Authors in [7] claim that integrating DFIG generation to the grid leads to a more stable inter-area mode. DFIG turbine system equipped with Voltage and power control loop provides better damping of low power oscillation [8]. Active and reactive power control of DFIG is achieved via the GSC and RSC [1]-[5]. In [11], two feedback control schemes are investigated using reactive power feedback and voltage feedback control. Besides, reactive power feedback control is provided faster system oscillation damping. Reactive power modulation is performed by adding a damping controller using the voltage magnitude as a reference signal [12]. Similarly, DFIG with a supplementary damping controller was investigated using two input signals to the active and reactive power control loop. Tie-line power and voltage phase angles are used for the supplementary damping controller [4]. This paper uses a dual POD controller to ensure the power system's resiliency during power oscillations due to load change or grid side fault. The generator's speed is considered a feedback signal to the reactive power POD, while the bus angle is a feedback signal for the active power POD.

II. SYSTEM MODELING AND CONTROL

Kundur's two-area four-machine system [13],[14] integrated with a DFIG-based wind farm, as depicted in Fig.1, is developed in PSCAD/EMTDC software to study the proposed

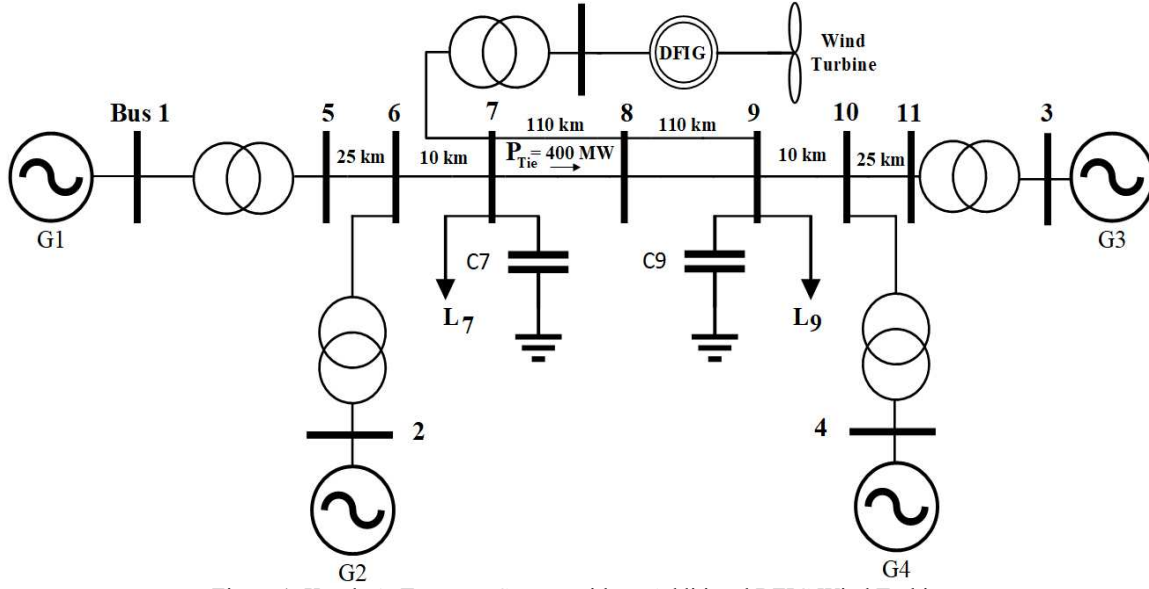


Figure 1: Kundur's Two-area System with an Additional DFIG Wind Turbine

dual POD controller effectiveness. The development of the model is performed in three stages. First, developing and testing the two-area system model in PSCAD/EMTDC. Next, a DFIG wind turbine system model was developed and integrated into the two-area system model. Finally, an oscillation-damping controller is integrated into the active and reactive power control loop of the DFIG model. Both areas I and II have two identical synchronous generators with a rated power of 900 MW, and they are equipped with turbine governors. PSS, seen in Fig. 2, is implemented at generator 1 and generator 3 with parameters listed in Tab. I. In addition, a standard IEEE ST1A excitation system model is used for each generating unit [15]. Turbine speed governors are developed for all four synchronous generator units, as seen in Fig. 3, with the parameters listed in Table II. Under steady-state conditions, the tie line is heavily loaded with a 400-MW power transfer from Area 1 to Area 2 across the weak transmission tie-line. The aggregated fixed load and shunt capacitors for reactive power supply are connected to buses 7 and 9. A single section lumped π model is used for all transmission lines, and the effect of the electromagnetic traveling wave is neglected. The wind farm is represented by one aggregated DFIG rated 180MW, connected to bus 1 in area I. The aerodynamic model of the wind turbine can be described in (1), and the total delivered by the wind turbine is represented (2), where R is the blade radius, ρ is the air density, v_{wind} is the wind speed; ω_r is the wind turbine angle speed, β is the blade pitch angle in degree, and λ is tip speed ration (3). C_p is the power coefficient which is defined as (4). The MPPT control continues to operate in the maximum turbine capacity region, as shown in Fig. 4 but limits the maximum electrical power command to be the turbine capacity. If the wind speed is above the rated wind speed, the input wind power will be greater than the output electrical power by increasing the pitch angle. Thus,

the pitch angle controller will reduce the captured wind power) and maintain the rotor speed at 1.3 P.U. The block diagram of the pitch-angle control is shown in Fig. 4 [12].

$$T_{aero} = \frac{0.5 \rho S v_{wind}^3 C_p(\beta, \lambda)}{\omega_r} \quad (1)$$

$$P_w = \frac{1}{2} \cdot \rho \cdot \pi \cdot R^2 \cdot V_w^3 \cdot c_p(\lambda, \beta) \quad (2)$$

$$\lambda = \frac{\omega_t \cdot R}{V_w} \quad (3)$$

$$c_p = 0.645 \left\{ 0.00912\lambda + \frac{-5 - 0.4(2.5 + \beta) + 116\lambda_i}{e^{21\lambda_i}} \right\} \quad (4)$$

$$P_s = -\frac{3}{2} v_{qs}^e i_{qs}^e = \frac{3}{2} \frac{L_m}{L_s} v_s i_{qr}^e \quad (5)$$

$$Q_s = -\frac{3}{2} v_{qs}^e i_{ds}^e = \frac{3}{2} \frac{L_m}{L_s} v_s \left(i_{dr}^e - \frac{v_s}{\omega_s L_m} \right) \quad (6)$$

$$P_g = \frac{3}{2} \frac{1}{K_T} (v_{ds} i_{dg} + v_{qs} i_{qg}) = \frac{3}{2} \frac{v_s}{K_T} i_{dg}^e \quad (7)$$

$$Q_g = \frac{3}{2} \frac{1}{K_T} (v_{qs} i_{dg} - v_{ds} i_{qg}) = -\frac{3}{2} \frac{v_s}{K_T} i_{qg}^e \quad (8)$$

Conventional vector control is adopted at the rotor side to independently regulate the active and reactive power of the stator. Under a steady-state condition, instantaneous active and reactive power outputs of the DFIG stator can be expressed in

(5) and (6), respectively. As seen, the active power of the stator output (torque) can be controlled by i_{qr}^e and the stator reactive power output can be controlled by i_{dr}^e . The representation of the Rotor Side Controller (RSC) can be illustrated in Fig. 5. Similarly, the instantaneous active and reactive power outputs from the GSC can be expressed in (7) and (8), respectively. Lastly, the GSC can be illustrated in Fig. 6.

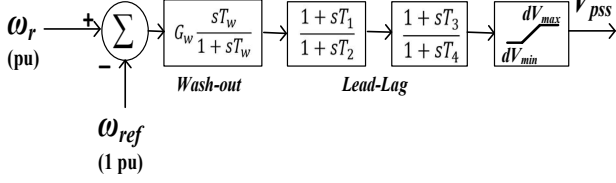


Figure 2: Power System Stabilizer Controller

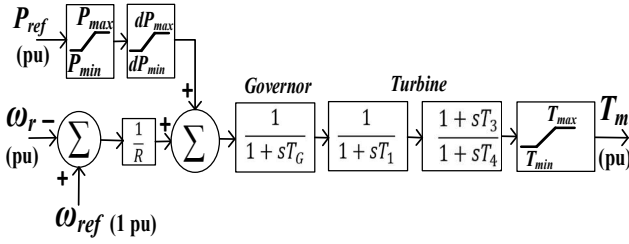


Figure 3: Turbine Speed Governor

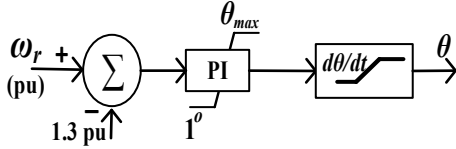


Figure 4: Wind Turbine Pitch Angle Controller

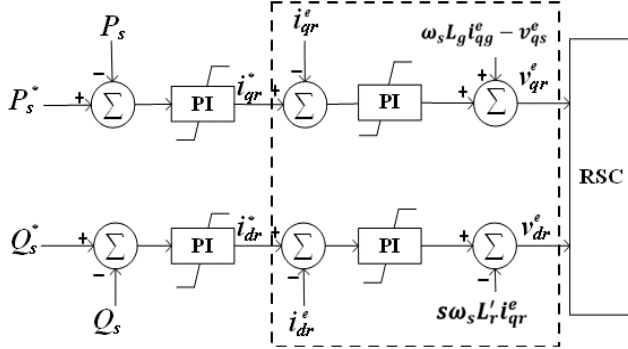


Figure 5: Rotor Side Vector Control

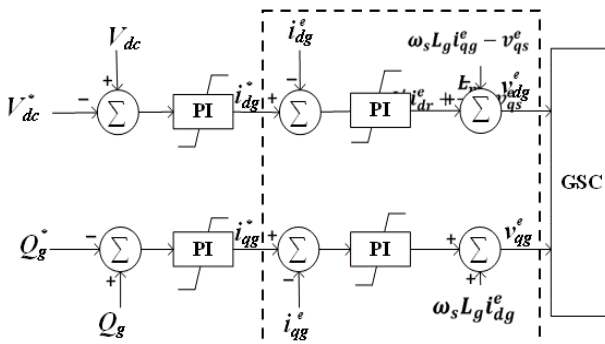


Figure 6: Grid Side Vector Control

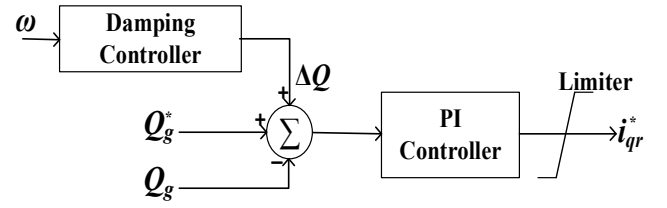


Figure 7: The Reactive Power POD Controller

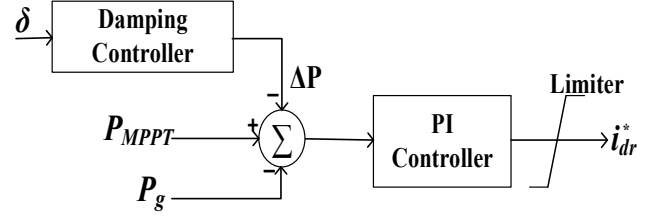


Figure 8: The Active Power POD Controller

TABLE I: PARAMETER SUMMARY OF WIND TURBINE MODEL

Quantity	Value
P_{rated} (MW)	3.6
V_{wind_rated} (m/s)	14
Inertia (s)	4.29

TABLE II: PARAMETER SUMMARY OF THE DFIG MODEL

Quantity	Value	Quantity	Value
MVA Rating	4	R_s (pu)	0.017
V_{rated} (kV)	4.16	R_r (pu)	0.015
Inertia (s)	0.9	L_m (pu)	4.4
S-to-R Turns Ratio	1	L_{ls} (pu)	0.179
V_{DC-Bus} (kV)	3	L_{lr} (pu)	0.156

III. INTEGRATION OF THE POWER OSCILLATION DAMPING CONTROLLER INTO THE RSC

The reactive POD (Q-POD) controller is integrated into the RSC control scheme, as shown in Fig. 7. The generator rotor speed is selected as a feedback control signal to activate the Reactive POD. The inertia caused by the large synchronous machines may delay the feedback in the control signal associated with the speed. Yet, the nearest synchronous generator speed is considered to omit the delays. The washout filter removes the steady-state component of the input signal, ω [7]. The reactive power reference for the damping controller, ΔQ , can be expressed in (9) where G_1 and T_{w1} are the gain and the time constant of the washout filter, respectively. In this paper, the value of T_{w1} is set at 10 sec. Similarly, the active power POD (P-POD) control scheme is added to the RSC controller, as shown in Fig. 8. Active power and bus angle (δ) are closely related. Thus, the Point of Common Coupling (PCC) bus angle is considered a feedback control signal for the active power control loop.

Using the remaining active power in the wind turbine, ΔP is obtained as in equation (10). where G_2 and T_{w2} are the gain and the time constant of the washout filter, respectively. Likewise, the washout filter of the reactive POD (T_{w1}), T_{w2} is also considered as 10 sec.

$$\Delta Q = \omega \times G_1 \times \frac{sT_{w1}}{1 + sT_{w1}} \quad (9)$$

$$\Delta P = \delta \times G_2 \times \frac{sT_{w2}}{1 + sT_{w2}} \quad (10)$$

IV. RESULTS

Time-domain Simulations on PSCAD are performed to test the effectiveness of the Q-POD, P-POD and Dual POD damping controllers, which are then compared with local PSS on both areas of the power system. The system operates under a steady state with rated power wind generation and the power transfer between the two areas at 400 MW. First, the two-area system is tested under a sudden load increase of 100MW on area II. The line power oscillation occurs immediately after the load increase by around 50 sec. It can be seen in Fig. 9 that higher oscillations exist when there is no damping controller in the system. To test the efficacy of the power oscillation damping of DFIG, it is compared to PSS on both areas of the power system. PSS is installed at generator 1 and generator 3 in the two-area power system.

The oscillation damping performance of PSS and the DFIG dual POD are illustrated in Fig. 10. In this case, the dual active POD of the DFIG provided better damping to the tie-line power compared to the PSS. Also, combining PSS and dual POD of DFIG yields much better damping to the tie-line power. The dual POD of the DFIG is evaluated under a three-phase to ground fault at bus 7 and compared with PSS and the dual POD case. As seen in Fig. 11, combining DFIG and PSS dual POD gives the best oscillation damping results. Generators' voltages response to the short circuit fault is tested at Gen. 1 and 3. The DFIG dual POD and PSS can improve the voltage response to reach the steady-state value faster with less oscillation, and less curtailment is observed at DFIG, as seen in Fig. 12. Lastly, the system stability is studied based on a frequency sweep using the impedance ratio. As shown in Fig. 1, the system is divided into two subsystems at bus 11 and then perturb through generator 3. The Nyquist Stability Criterion (NSC) is applied to the system's open-loop gain to determine the system stability. As seen in Fig. 13, the system is stable with the dual POD and PSS as the NSC does not encircle $(-1, j0)$ [16], [17].

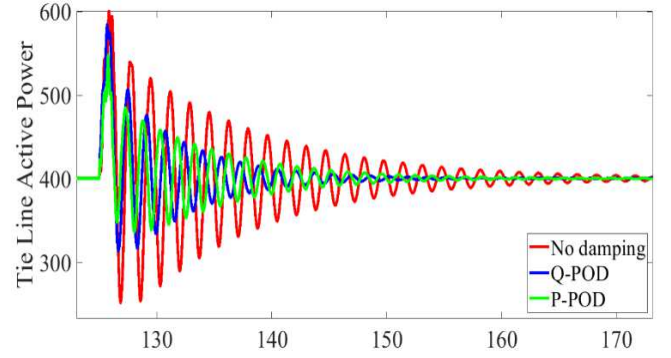


Figure 9: P-POD and Q-POD Comparison under a Load Increase on Area II

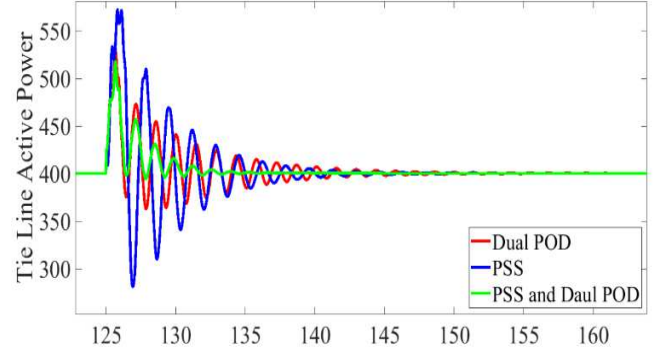


Figure 10: PSS and Dual POD Comparison under a Load Increase on Area II

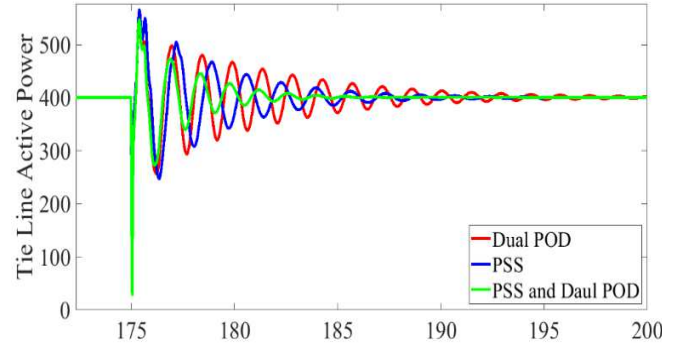
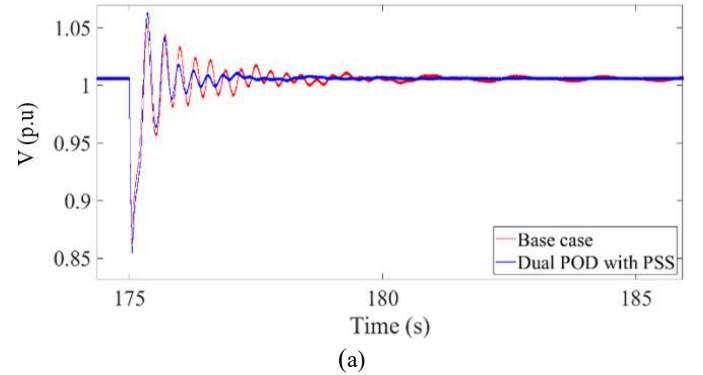


Figure 11: PSS and Dual POD Comparison under a Three-Phase-to-Ground Fault on Bus 7



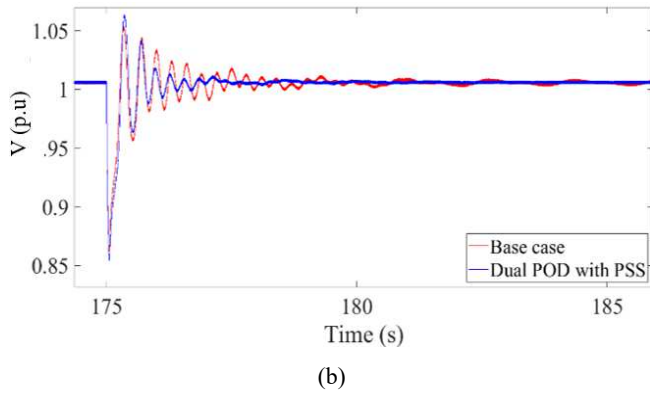


Figure 12: Dynamic Voltage Response under a Three-Phase-to-Ground Fault at Gen. 1 side (a) and Gen. 3 Side (b)

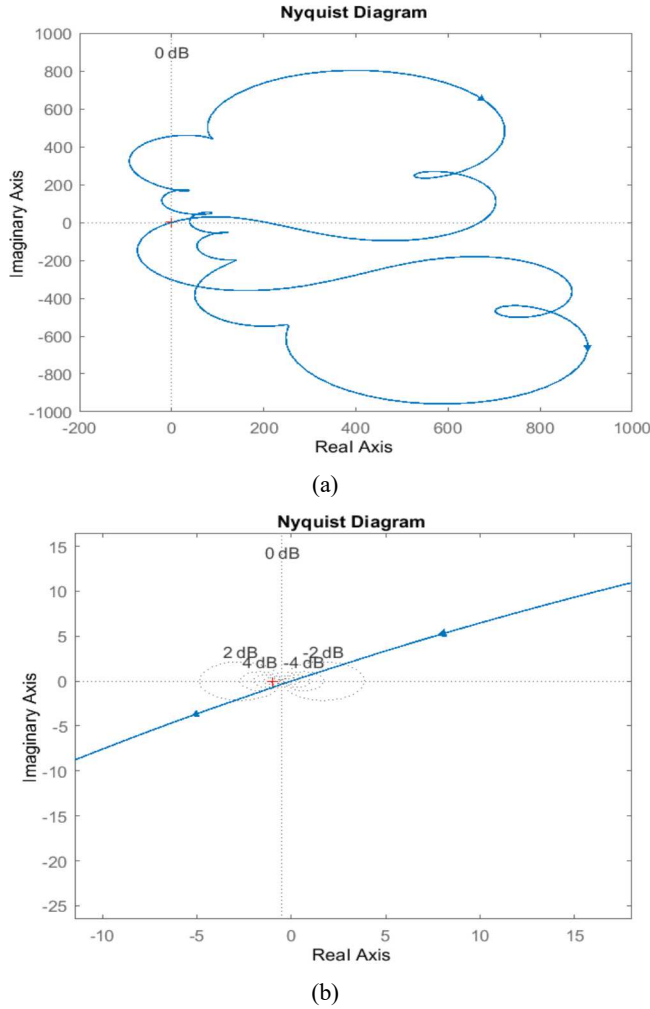


Figure 13: Nyquist Diagram of the Two Area System with Dual POD and PSS Integration (a) Zoomed-in (b)

V. CONCLUSION

This paper uses active and reactive power damping controller methods for damping interarea oscillation using a DFIG wind farm. In addition, applying dual POD to the power system oscillation yields better results that are close to having PSS on both power system areas. Combining dual POD with PSS provided promising results, especially in reducing the mechanical torque oscillation in wind turbine generation

systems. Moreover, an optimal selection of wind generators to be equipped with dual POD functionality for best interarea oscillation damping in an extensive power system with high wind generation penetration. Lastly, a stability analysis is performed to investigate the effects of the dual POD.

VI. REFERENCES

- [1] M. Wang et al., "Impedance Modeling and Stability Analysis of DFIG Wind Farm with LCC-HVDC Transmission," in *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, Mar, 2022.
- [2] S. Das and B. Singh, "Enhanced Control of DFIG Based Wind Energy Conversion System under Unbalanced Grid Voltages Using Mixed Generalized Integrator," in *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, Mar, 2022.
- [3] N. Gurung, R. Bhattarai and S. Kamalasan, "Optimal Oscillation Damping Controller Design for Large-Scale Wind Integrated Power Grid," in *IEEE Transactions on Industry Applications*, vol. 56, no. 4, pp. 4225-4235, July-Aug. 2020.
- [4] W. Du, J. Bi, T. Wang and H. Wang, "Impact of grid connection of large-scale wind farms on power system small-signal angular stability," in *CSEE Journal of Power and Energy Systems*, vol. 1, no. 2, pp. 83-89, June 2015.
- [5] N. Mithulananthan, C. A. Canizares, J. Reeve and G. J. Rogers, "Comparison of PSS, SVC, and STATCOM controllers for damping power system oscillations," in *IEEE Transactions on Power Systems*, vol. 18, no. 2, pp. 786-792, May 2003.
- [6] S. Isik, H. Nath, M. Alharbi and S. Bhattacharya, "Negative Sequence Component Elimination with M-STATCOM for Versatile FACTS," *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*, 2019, pp. 7069-7073, doi: 10.1109/IECON.2019.8927805.
- [7] J. J. Sanchez Gasca, N. W. Miller, and W. W. Price, "A modal analysis of a two-area system with significant wind power penetration", in *Proc. 2004 IEEE PES Power Systems Conf. Expo*.
- [8] N. J. Olimpo Anaya-Lara, F. Michael Hughes and G. Strbac, Influence of windfarms on power system dynamic and transient stability, *Wind Engineering*, vol. 30, pp. 107127, 2006.
- [9] D. Gautam, V. Vittal, and T. Harbour, "Impact of increased penetration of DFIG-based wind turbine generators on transient and small signal stability of power systems", *IEEE Trans. Power Syst.*, vol. 24, no. 3, pp. 1426-1434, Aug, 2009.
- [10] F. M. Hughes, O. A. Lara, N. Jenkins, G. Strbac, "A power system stabilizer for DFIG-based wind generation", *IEEE Trans. Power Syst.*, vol. 21, no. 2, pp. 763772, May 2006.
- [11] J. Ma, L. Wang and P. Cheng, "An Active Damping Control Strategy for Grid-Tied Type-4 Wind Turbine Generator Based on Energy Feedback," in *IEEE Transactions on Power Electronics*, vol. 37, no. 5, pp. 5003-5020, May 2022.
- [12] M. Klein, G. J. Rogers, S. Moorthy and P. Kundur, "Analytical investigation of factors influencing power system stabilizers performance," in *IEEE Transactions on Energy Conversion*, vol. 7, no. 3, pp. 382-390, Sept. 1992.
- [13] J. Liang, "Wind Energy and Power System Interconnection, Control, And Operation For High Penetration Of Wind Power," in *PhD Dissertation*, Atlanta, Georgia Institute of Technology, 2012.
- [14] J. K. Hwang and J. Shin, "Identification of Interarea Modes From Ambient Data of Phasor Measurement Units Using an Autoregressive Exogenous Model," in *IEEE Access*, vol. 9, pp. 45695-45705, 2021, doi: 10.1109/ACCESS.2021.3067213.
- [15] A. A. Alsakati, C. A. Vaithilingam, J. Alnasseir and A. Jagadeeshwaran, "Simplex Search Method Driven Design for Transient Stability Enhancement in Wind Energy Integrated Power System Using Multi-Band PSS4C," in *IEEE Access*, vol. 9, pp. 83913-83928, 2021, doi: 10.1109/ACCESS.2021.3085976.

- [16] S. Isik and S. Bhattacharya, "Reliability and Cost Modeling of a Modular Multilevel Converter," 2021 IEEE 12th International Symposium on Power Electronics for Distributed Generation Systems (PEDG), 2021, pp. 1-8, doi: 10.1109/PEDG51384.2021.9494184.
- [17] Alharbi, M.; Isik, S.; Alkuhayli, A.; Bhattacharya, S. Power Ripple Control Method for Modular Multilevel Converter under Grid Imbalances. *Energies* 2022, *15*, 3535. <https://doi.org/10.3390/en15103535>